

Sec. 3.6: Cartesian products

Definition.

- An **ordered pair** of items is written .
- The **Cartesian product** of two sets A and B is the set:
- $(a_1, b_1) = (a_2, b_2) \leftrightarrow$
- In general, $A_1 \times A_2 \times \cdots \times A_n =$

Example 1. Let $A = \mathbb{R}$. Then $A \times A =$

Example 2. Let $A = \{a, b\}$, $B = \{x, y, z\}$. Write each of the following set using roster notation.

1. $A \times B =$

2. $B \times A =$

Exercise 3. Let $A = \{1, 2\}$, $B = \{2, 3\}$, $C = \{+, \$\}$.

1. $(A \cap B) \times C =$

2. $B \times C \times A =$

Definition.

- A sequence of characters is called a **string**.
- **String notation:** If A is a set of symbols or characters, the elements in A^n can be written without the usual punctuation.

Example 4. Let $A = \{0, 1\}$, then $A^2 = ?$

- using n -tuples notation:
- using string notation:

Definition.

- The **empty string** is the unique string whose length is , denoted by .
- If s and t are two strings, then the **concatenation** of s and t (denoted st) is the string obtained by putting s and t together.

Example 5. Let $s = 01$, $t = 101$. Then

- $st =$
- $t1 =$
- $\lambda s =$

Exercise 6. Give an element of each of the following set in string notation.

- $\{xy : x \in \{0, 1\}^2, y \in \{a, b, c\}^3\}$
- $\{10t : t \in \{a, b\} \cup \{a, b\}^2 \cup \{a, b\}^3\}$

Summary - fill in the blanks. Answers will be posted on Canvas.

- $A \times B$ denotes the (name) of sets A and B ,
Definition in set builder notation: $A \times B =$
- For example, if $A = \{1, 2\}$, $B = \{2, 3\}$, then $A \times B =$.
- $A_1 \times A_2 \times \cdots \times A_n =$.
The elements in this set are called .
- If $A = \{1, 2\}$, $B = \{2, 3\}$, express the elements of $A \times B$ in string notation:
 $A \times B = \{$ $\}$.
- Let $x = 112$, $y = 212$ be two strings from the set $\{1, 2\}^3$,
then the concatenation of x and y is